

Customer Churn Prediction

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This project will focus on finding out the factors which affect customer churn and developing Machine Learning model that will predict whether the customer is likely to leave the company's services using demographics, account information, and service-related features.

Business Problem

Customer churn is one of the major problems that companies face because losing customers can lose income and increase the cost of acquiring new customers. This project aims to identify what causes customer churn and build a Machine Learning Model to predict whether a customer likely to leave a company's services or not.

Research Questions

1. Which customer characteristics have the greatest influence on customer churn?
2. What is the effect of tenure, contract type, and monthly charges on customer churn?
3. Which machine learning model provides the best prediction performance?

Background/History

The ability to retain customers is an important aspect for businesses that operate on a subscription model since customers have the option to switch services if their demands with regards to prices, services or contracts are not satisfied. Churn analysis uses historical customer data to identify patterns that differentiate customers between those who remain and who leave. Machine learning classification models can support this process by estimating churn risk and helping a business focus retention efforts on customers who may need additional attention.

Data Explanation

The analysis uses the Telco Customer Churn dataset. The original dataset contains 7,043 customer records and 21 variables. The variables include customer demographics, tenure, phone

and internet services, online security and support services, contract type, paperless billing, payment method, monthly charges, total charges, and the target variable Churn. The customerID field is an identifier which has been dropped for the modeling purpose. The initial missing-value check did not show standard null values, but TotalCharges was stored as an object because a small number of records had blank values. TotalCharges was converted to numeric, the blank records were treated as missing and removed. The cleaned dataset contained 7,032 records. All categorical variables were converted to numeric indicator variables through one-hot encoding, and Churn was recorded as 0 for No and 1 for Yes.

Methods

The analysis was performed in Python. Exploratory data analysis was used to examine the churn distribution and relationship with contract type, tenure, monthly charges, and internet service. Three different models, Logistic Regression, Decision Tree, and Random Forest classification models were developed using an 80/20 train/test split. The Models were compared by using accuracy, precision, recall, and F1-score. Five-fold cross-validation and a small Random Forest hyperparameter search were also used to strengthen model evaluation.

Analysis

The cleaned dataset had 7,032 records. On average, 26.58% of customers churned. The highest churn ratio was recorded by the month-to-month customers at 42.71%, followed by one year customers at 11.28%, and finally two year customers at 2.85%. Churned customers have shorter average tenure (17.98 months) than customers who remained (37.65 months), and their monthly average monthly charges were higher (\$74.44 compared to \$61.31) than non-churned customers. Moreover, fiber optic customers also had a higher churn rate (41.89%) than DSL customers (19.00%).

Logistic Regression has performed most effectively, with accuracy of 0.805, precision of 0.650, recall of 0.575, and an F1-score of 0.610. Random Forest achieved 0.787 accuracy, while Decision Tree achieved 0.719. Five-fold cross-validation produced average accuracy of 0.804 for Logistic Regression, 0.793 for Random Forest, and 0.732 for Decision Tree. Random Forest tuning selected 100 estimators and a maximum depth of 10 and improved its cross-validation accuracy from 0.793 to 0.803. Important Random Forest features included tenure, TotalCharges, MonthlyCharges, contract type, and fiber-optic internet service.

Visualization

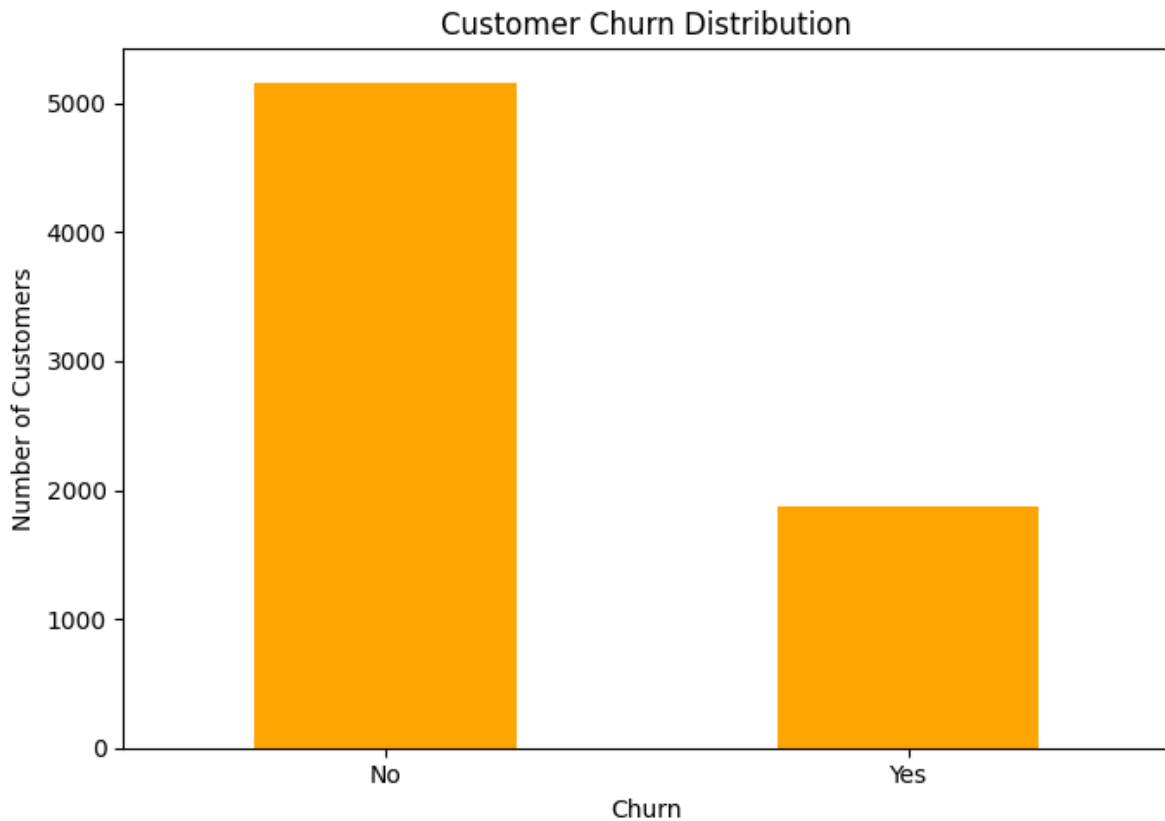


Figure 1. Customer Churn Distribution

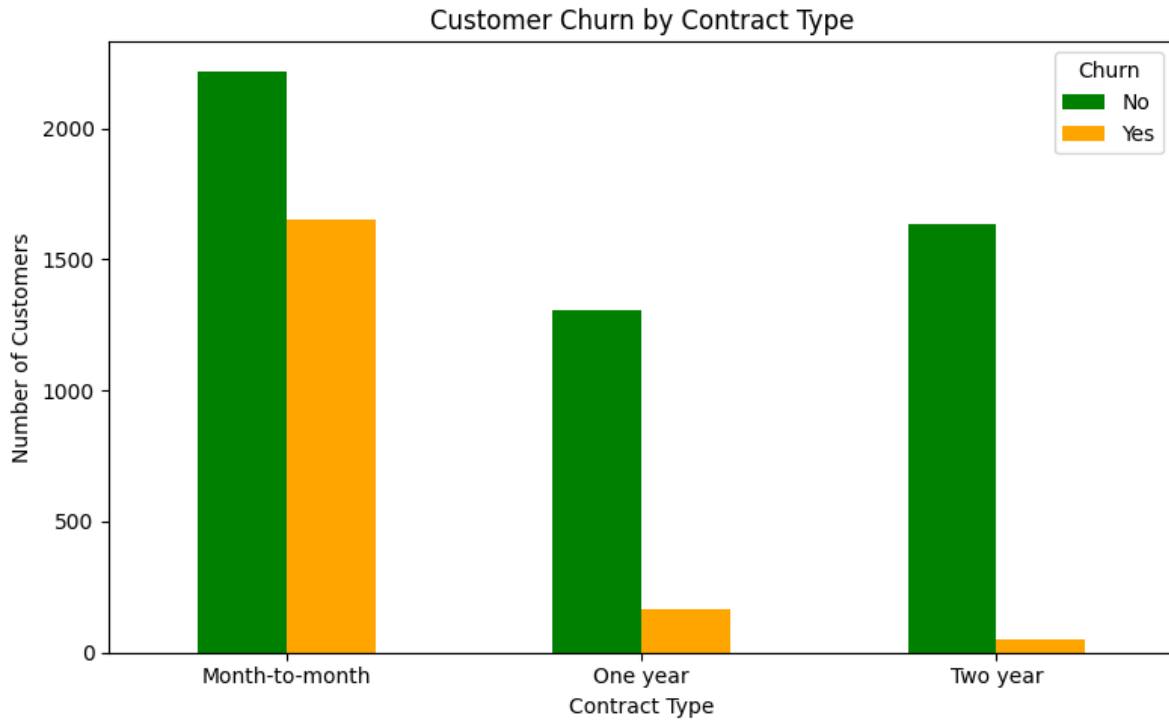


Figure 2. Customer Churn by Contract Type

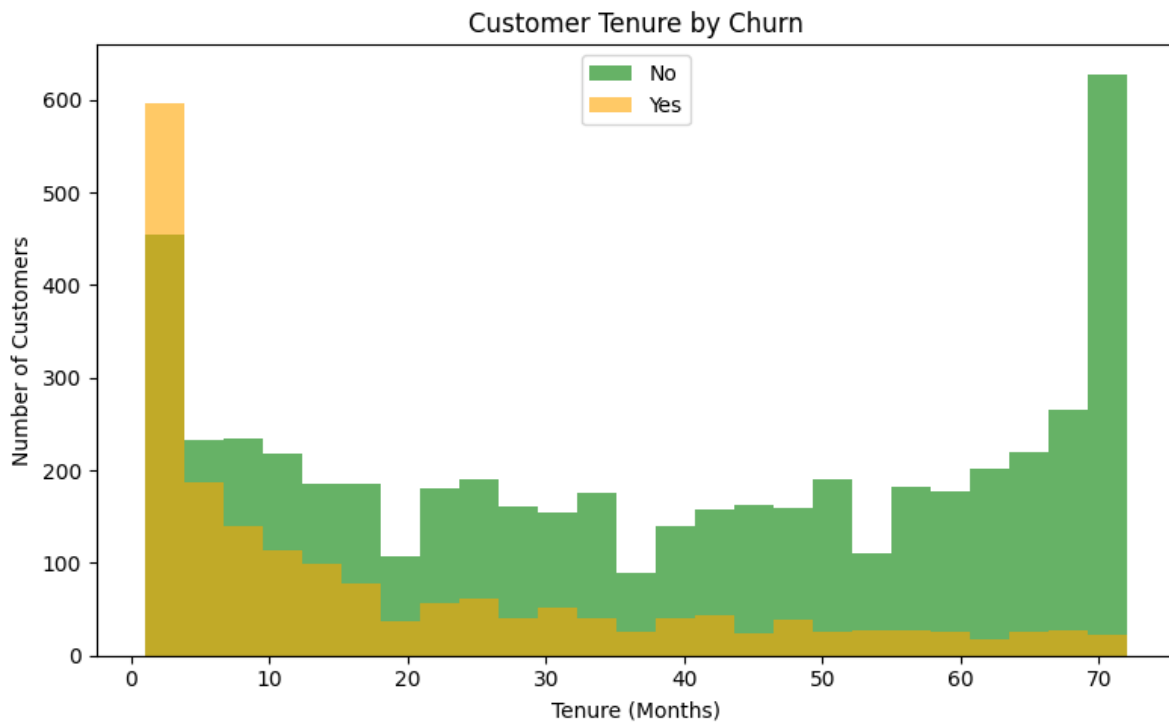


Figure 3. Customer Tenure by Churn

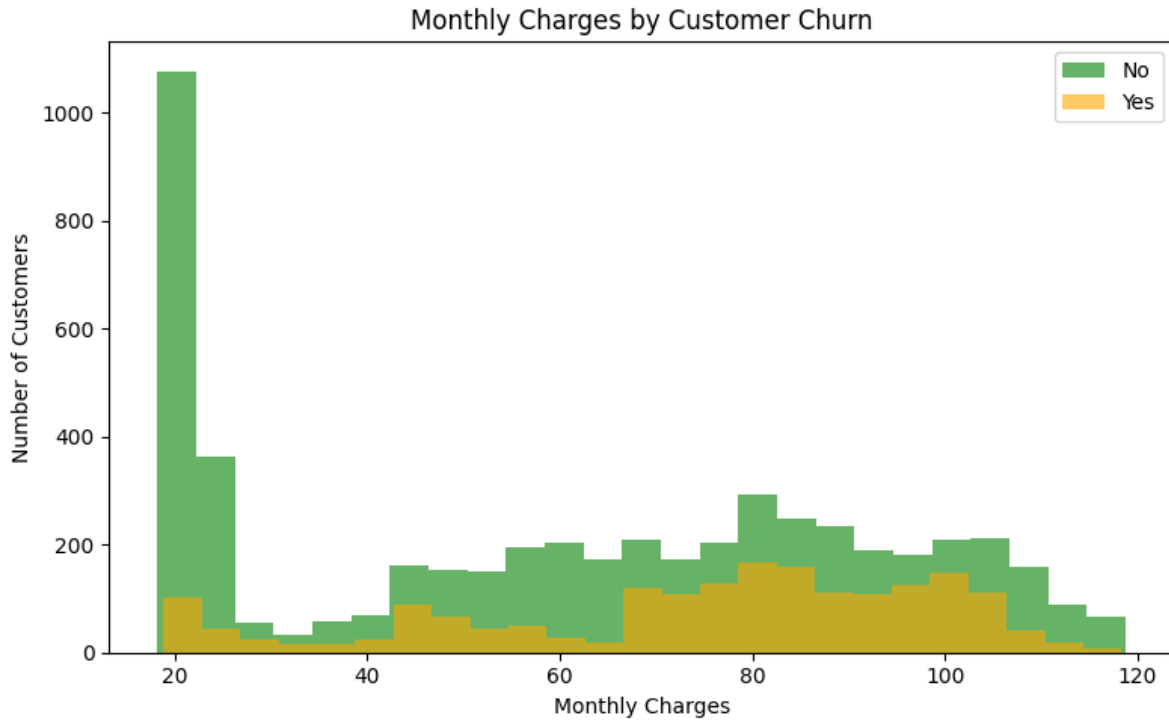


Figure 4. Monthly Charges by Customer Churn

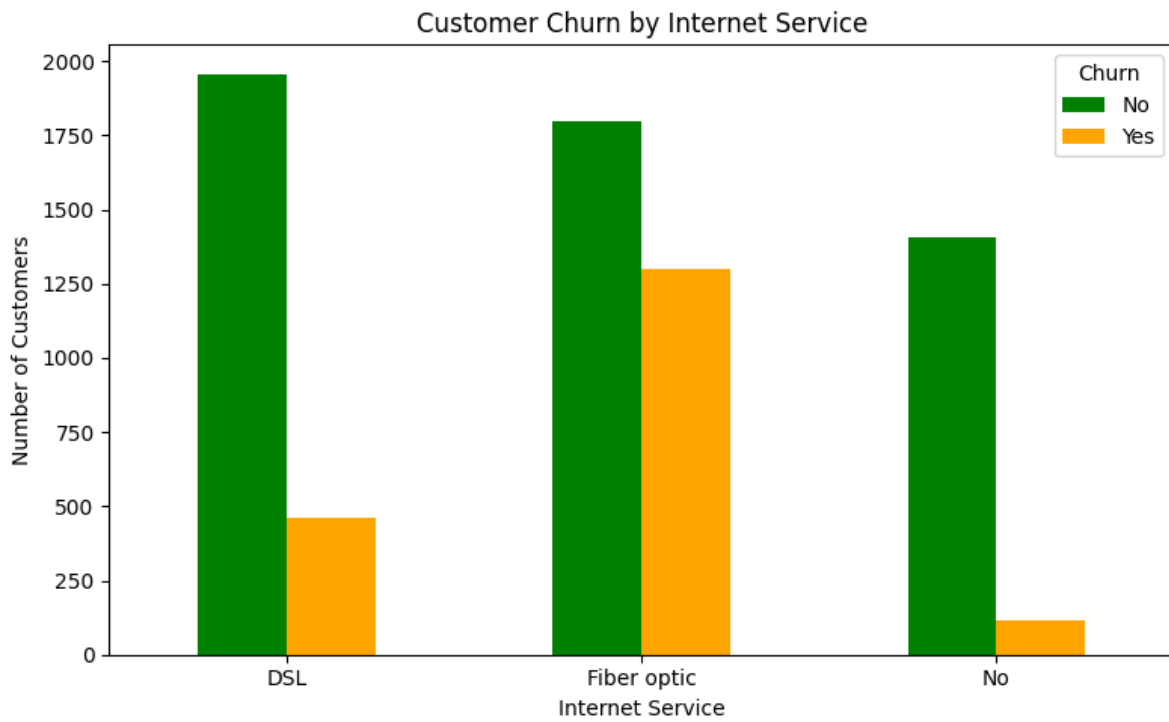


Figure 5. Customer Churn by Internet Service

Table 1. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.805	0.650	0.575	0.610
Decision Tree	0.719	0.470	0.463	0.466
Random Forest	0.787	0.622	0.511	0.561

Conclusion

The results show that contract type, tenure, monthly charges, and internet service are associated with customer churn. Logistic Regression performed best among the three models with 0.805 test accuracy and 0.804 cross-validation accuracy. Random Forest improved after tuning but did not outperform Logistic Regression. Overall, the analysis indicates that machine learning can help identify customers who may be at greater risk of leaving and support customer-retention decisions.

Assumptions

It is assumed that the customer records are accurate and that the Churn variable correctly identifies whether a customer left the service. It is also assumed that the correlation between variables that has been detected on the basis of historical customer data will be helpful for predicting churn for similar customers.

Limitations

The dataset is a sample dataset and does not refer a specific real telecommunications industry. Hence, the findings should not be treated as direct evidence about a particular company's current customers. The dataset also does not include factors such as customer satisfaction levels, service outages, competitor offers, complaint history, or detailed geographic

information, that may affect churn. In addition, the churn rates are imbalanced because substantially fewer customers churned than remained.

Challenges

The main difficulties will be transforming categorical variables to a form that classification models can use, handling empty values of TotalCharges, evaluation models when the target classes are not evenly distributed, and balancing between prediction accuracy and interpretability of models.

Future Uses/Additional Applications

This churn prediction model can be used to support a customer retention campaigns, identify customers for targeted offers, follow up on customer service issues, and help managers to understand patterns associated with customer loss. Similar approach can also be used in subscription based industries like insurance, banking, online entertainment, and memberships services.

Recommendations

Potential future work could include exploring additional classification models, advance feature engineering, and approaches for handling class imbalance. The model should also be validated using additional or organization specific customer data before being used for business decisions.

Ethical Assessment

The dataset includes customer demographic and service-related information, so predictions should be used carefully. The churn prediction model must help make retention decision-making easier instead of deciding how the customer is going to be handled. The organization should evaluate the impact of the prediction on the creation of any disparities

between the groups of customers and must avoid using sensitive characteristics that could result in any discriminatory practices. The customers must not be denied service or opportunities because a model assign them a higher churn risk.

Appendix

Dataset: Telco Customer Churn. The original dataset has 7,043 records and 21 variables. After converting TotalCharges to numeric and removing TotalCharges blank values, 7,032 records were used for modeling.

Selected Data Dictionary

Variable	Description
CustomerID	Unique customer identifier; removed before modeling.
Tenure	Number of months the customer has stayed with the company.
InternetService	Customer internet service type: DSL, fiber optic, or no internet service.
Contract	Customer contract type: month-to-month, one year, or two year.
PaymentMethod	Method used by the customer to make payments.
MonthlyCharges	Amount charged to the customer each month.
TotalCharges	Total amount charged to the customer.
Churn	Target variable indicating whether the customer left the service.

References

Kaggle. (n.d.). Telco customer churn. <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). An introduction to statistical learning: With applications in Python. Springer.